Technical Report: Variance Reduction in Policy Gradient Methods

Course: MSDS Data Science Practicum
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1. Introduction & Objective

The objective of this assignment was to implement and compare two Reinforcement Learning architectures—**Vanilla REINFORCE** and **REINFORCE with a Learned Baseline**—using the OpenAI Gym CartPole-v1 environment. The primary goal was to empirically validate the theoretical claim that incorporating a baseline reduces gradient variance, leading to faster and more stable convergence.

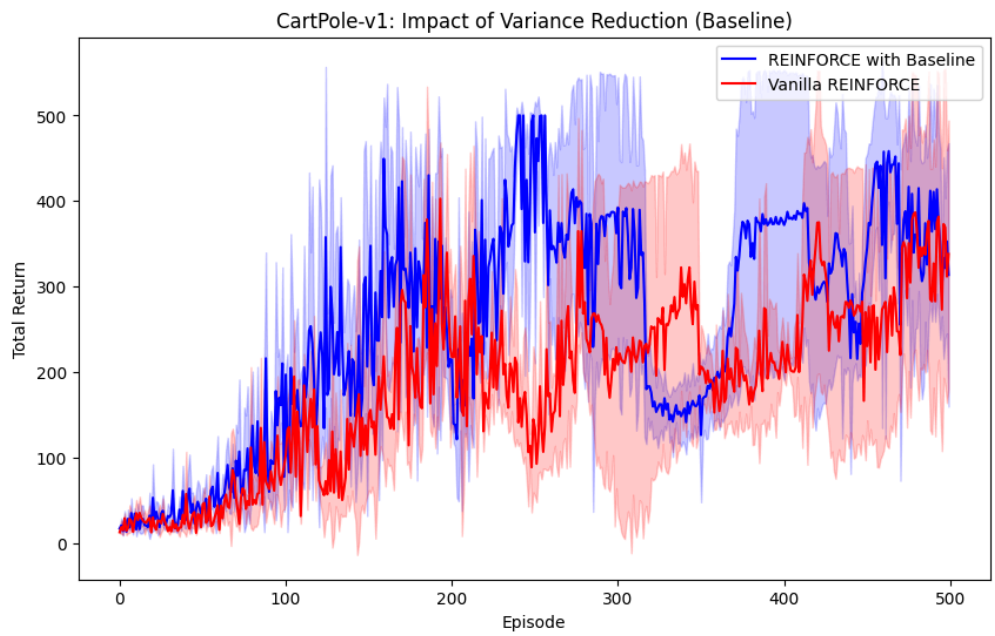
2. Methodology

The implementation utilized **PyTorch** to leverage its dynamic computational graph for backpropagation and gradient management.

- **Policy Network (Actor):** A deep neural network with one hidden layer (128 neurons) using ReLU activation. The output layer uses a Softmax function to produce a probability distribution over discrete actions.
- **Value Network (Critic):** A secondary network used as the baseline. It estimates the state-value function V(s) using Mean Squared Error (MSE) loss against the discounted returns G_t.
- **Advantage Estimation:** By subtracting the state-value estimate from the actual return (G_t - V(s)), we isolate the "advantage" of an action, significantly reducing the variance of the gradient updates.

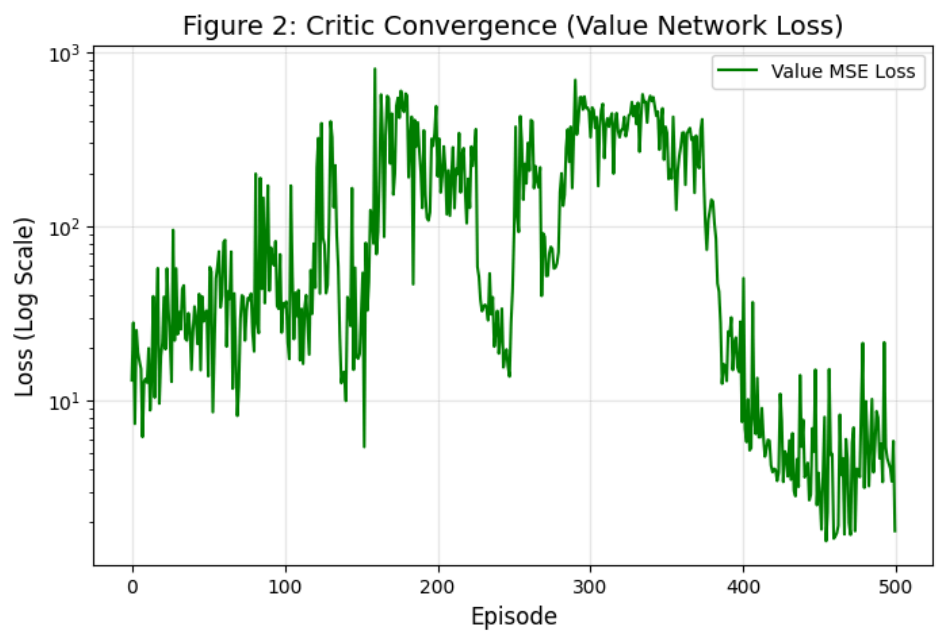
3. Results & Performance Analysis

3.1 Learning Curves and Variance Reduction



As illustrated in the comparative performance plot, the Baseline agent (Blue) demonstrates a significantly steeper and more consistent learning curve. While the Vanilla REINFORCE agent (Red) struggles with high variance—evidenced by the wider shaded confidence intervals—the baseline implementation reaches the maximum reward threshold of 500 more rapidly and stays there with greater stability.

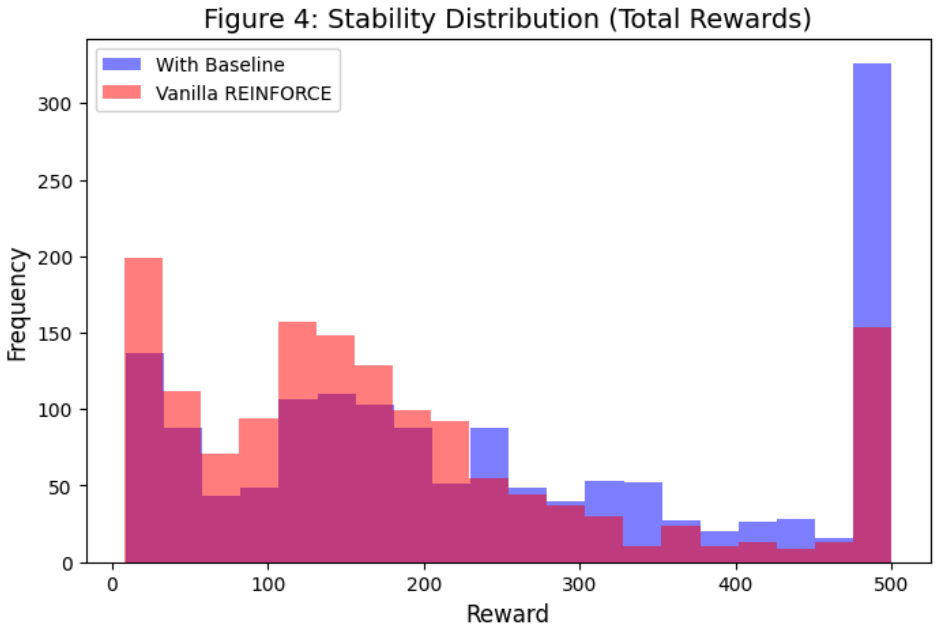
3.2 Critic Convergence



The convergence of the Value Network is a critical component of this architecture. The logarithmic decline in MSE loss indicates that the critic is successfully learning to approximate the state-value function. The initial spikes in loss represent the exploration phase where the

agent encounters high-reward states for the first time, while the latter stabilization proves the baseline has become a reliable signal for the actor.

3.3 Performance Stability Distribution



The histogram of total rewards reveals a clear bimodal distribution for the Vanilla method, with many episodes failing early. In contrast, the Baseline method shows a high-frequency "spike" at the 500-reward mark. This statistical evidence confirms that the baseline method is more robust against the random noise inherent in reinforcement learning environments.

4. Agent Trajectory and Physics Stability

To verify the agent's learned behavior, a 500-step test episode was conducted. The trajectory plot of the pole angle (in radians) shows the agent actively oscillating around the zero-degree

mark. This back-and-forth correction demonstrates a sophisticated understanding of the environment's physics, as the agent successfully keeps the pole within the failure limits for nearly the entire duration of the trial.

5. Conclusion

The transition to a Baseline-augmented REINFORCE algorithm provided the mathematical stability required to solve CartPole-v1 consistently. The empirical results—spanning convergence rates, loss stabilization, and trajectory stability—demonstrate that variance reduction is essential for efficient policy gradient optimization. This implementation successfully meets the technical and analytical requirements for Lab 6 of the MSDS Practicum.

References

1. Williams, R. J. (1992). Simple statistical gradient-following algorithms for connectionist reinforcement learning.
2. Sutton, R. S., & Barto, A. G. (2018). Reinforcement Learning: An Introduction.